## CHALLENGE #8

1. **Download Dataset**

   The dataset is taken directly from Kaggle (using the Kaggle API account) via Google Collab, using the Python programming language.

2. **Create a campaign performance report**

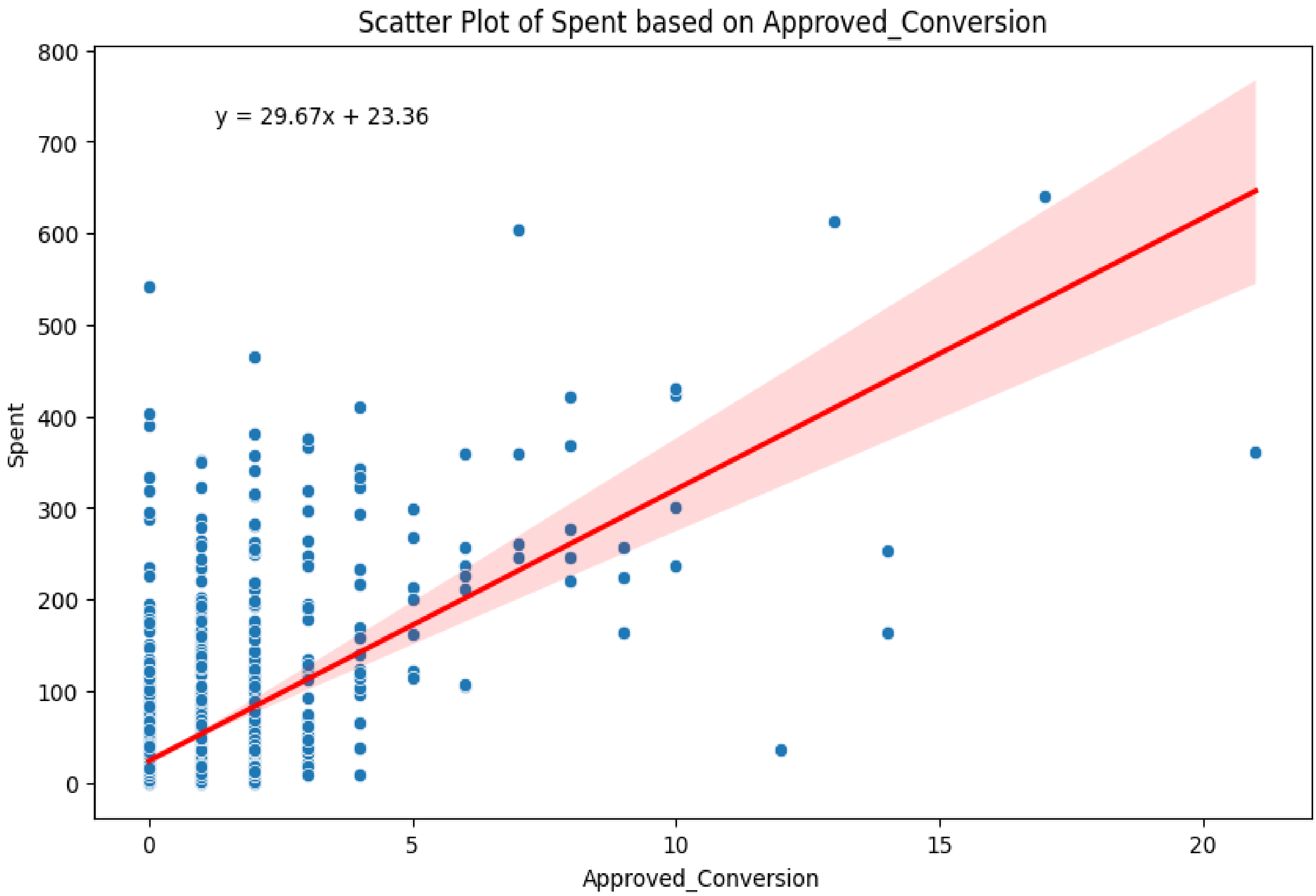
   - The overall average cost per approved conversion is **33.90%**

     Build A simple linear regression model to predict approved conversion based on money spent.

     How much will each approved conversion cost?

     To create a linear regression model in python we can use the library from sklearn. The independent variable is Approved_Conversion and the dependent variable is Spent.

     The two parameter values that will determine this model are regression coefficient and regression intercept. After the data is trained into a linear regression model, we get the regression coefficient of **29.66674802** and the regression intercept of **23.35503836**.




By using regression coefficient and regression intercept, we can predict spent based on approved conversion with the following formula

***Approved Conversion = (Regression coefficient × Spent) + Regression Intercept***

*Approved Conversion = (29.66674802 × Spent) + 23.35503836*

So it is found that every approved conversion cost of **±53.02 AUD**.

- Why might this differ from the average cost calculated above?
  The values of the two are different because the average uses a simple calculation that only divides the sum of all variable values by the number of observations without paying attention to the relationship between the two. In contrast, linear regression pays close attention to the relationship between the two. The conclusion is that averages provide an overview of the centre of the data distribution, while linear regression describes the relationship between two variables.

- Which of the values might be more representative of predictions?
  The more representative answer is the value of the linear regression model. A linear regression model is used when we want to understand the relationship between two variables and how one variable affects the other. It is very useful for prediction and causal analysis.